

Project Causal Machine Learning

Introduction

The data consist of 9,915 observations drawn from the 1991 Survey of Income and Program Participation (SIPP): each observational unit is a person aged 25–64 (plus spouse if present) in a household where at least one adult is employed and none are self-employed. This same SIPP sample was used by Chernozhukov et al. (2004, *Rev. Econ. Stat.*) to study the wealth impact of 401(k) participation via instrumental quantile regression and by Belloni et al. (2014, "Program Evaluation with High-Dimensional Data") for high-dimensional program-evaluation methods.

Like in both studies mentioned we will use net financial assets `net_tfa` as a principal outcome when estimating the effect of 401(k) eligibility `e401` on household wealth. Throughout, we refer to households **eligible** for a 401(k) as the **treated** group and those **not eligible** as the **untreated** group, net financial assets will be referred to as the **outcome**.

Identification Strategy

In reasoning about whether treated and untreated individuals (eligible for 401k) are conditionally exchangeable on outcome when correcting for income we make the following assumptions.

When talking about saving "preference", this is a character trait that is not necessarily dependent on income in the sense that there are plenty individuals with high income that have no tendency so save money. On the other hand one can make a better case that having a preference for saving is proxy for making sound life decisions and having a certain amount of intelligence and knowledge about the world. These traits can lead to making better choices in life and also appreciating the importance of accumulating savings. These same traits can lead to choosing higher education and jobs with larger income.

Income of course also affects the amount of assets one accumulates, savers accumulate more. Since high income jobs correlate with more robust employers there is a higher chance of these employers offering a 401k. So income affects eligibility in that sense.

The following shorthand parameters and DAG (Figure 1) are used to reason about exchangeability:

- Income (**I**)
- Treatment variable: Eligibility (**A**)
- Outcome variable: Accumulated financial assets (**Y**)
- (Unmeasured) preference for saving (**S**)

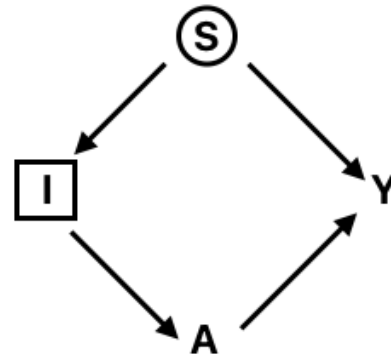


Figure 1. Directed acyclic graph (DAG) illustrating the assumed data-generating process for 401(k) eligibility and net financial assets.

Eligibility affects accumulated assets directly but also has a spurious association via the path $A \rightarrow I \rightarrow Y$. This is not a causal path but can bias the results when investigating the effect of eligibility on net financial assets. If this simple DAG represents the actual data generating process, by conditioning on income we are blocking this indirect path leaving only the direct path from eligibility to asset accumulation.

Conditioning on I therefore blocks the path $A \rightarrow I \rightarrow Y$, making Eligibility (A) conditionally exchangeable in regards to net financial assets (Y) when correcting for income (I)

$$Y^a \perp\!\!\!\perp A \mid I, L,$$

other than

- `e401` (Eligibility for 401(k)) (\$A\$)
- `net_tfa` (Net financial assets (in US \$)) (Y)

L denotes the remaining covariates "age", "inc", "fsize", "educ", "marr", "twoearn", "db", "pira", "hown".

It is however naive to assume that income will capture all of the non-direct association between treatment and outcome. Other employer-level factors that were not accounted for here and are independent of income could still influence both the probability of offering a 401(k) and workers' ability to accumulate liquid assets.

Baseline unadjusted linear regression

To set a baseline a simple unadjusted linear regression was performed of the outcome on the treatment. The unadjusted treatment effect is **19 559\$ (95% CI 17 000\$–22 119\$)**; see **Table 1**.

ATE and ATT estimation

With `e401` as treatment, `net_tfa` as outcome and the remaining above mentioned covariates a more robust estimation of the ATE was calculated using an AIPW and TMLE estimator.

To balance **simple parametric learners** with **flexible, non-parametric models**, the SuperLearner library throughout the analysis was set to:

```
"SL.glm", "SL.glmnet", "SL.glm.interaction", "SL.mean", "SL.ranger", "SL.earth"
```

G-computation and Inverse Probability Weights

As a basis for the AIPW and TMLE estimators, **counterfactual outcomes** Y^1 and Y^0 were predicted using g-computation and a SuperLearner ensemble model trained on outcome, treatment and covariates.

To obtain the propensity scores, another ensemble model using the same SuperLearner library was trained on the treatment and covariates.

The non-zero model weights of the ensemble models trained on the outcome and the propensity scores are summarized in **Table 2** and **Table 3**.

AIPW estimator of the ATE

Using a manually calculated textbook AIPW estimator for the ATE based on our previously calculated outcome counterfactuals and propensity scores, we obtain an **ATE estimate of 8 761 \$ (95% CI 6 853\$ –10 670\$)**; see **Table 4**.

TMLE estimator of the ATE

We repeated the calculation for the ATE using the textbook TMLE. Since our outcome is continuous we used linear regression to obtain our fluctuation model delta and updated our initial counterfactual accordingly. The **TMLE gives and ATE estimate of 8 481 \$ (95% CI 6 573 \$–10 390 \$)**; see **Table 5**.

Extreme variability in the outcome distribution and violation of positivity assumption

The **outcome** distribution suffers from **extreme outliers** as demonstrated in **Figure 2** and **Table 6**.

Kernel densities of the propensity scores for treated and untreated groups show little overlap, and the inverse weights exceed 20 for some of the untreated → **the positivity assumption is violated for the full population**.

Figure 3 visually represents the distributions of the propensity scores.

The absence of cross-fitting—coupled with heavy-tailed outcomes and possible positivity violations—is likely to skew the point estimates and inflate their standard errors.

To mitigate these issues due to positivity violation, we next focused on estimating the **Average Treatment Effect on the Treated (ATT)**, as the covariate distributions within the treated subgroup appear to be more balanced.

AIPW estimator of the ATT

We estimate the ATT using the AIPW approach. First, we build a model to predict the outcome for the untreated. We then compare each treated individual's actual outcome with their untreated counterfactual predicted from our model (that was based on the untreated).

To adjust for confounding, we also use the propensity score to reweight and correct the outcomes of untreated individuals. The AIPW yields an ATT estimate **10 548 \$ (95% CI 7 813 \$ –13 283 \$)** (see **Table 7**).

TMLE estimator of the ATT

To estimate the ATT, we use TMLE. First, we predicted the outcome under no treatment using only the untreated individuals. Then, we applied a targeting step that adjusted these predictions using information from both treated and untreated groups. Finally, we computed the ATT by comparing the observed outcomes in the treated group to their targeted counterfactual predictions. We also derived an influence function to estimate the standard error and construct a confidence interval. The TMLE gives an ATT estimate of **10 468 \$ (95% CI 7 720 \$ –13 216 \$)** (see **Table 8**).

Our estimations for the ATT deviate largely from the analysis performed by Chernozhukov and Belloni. This can be explained by the lack of cross fitting the estimators and performing no correction for the heavy-tailed outcome distribution.

Residual Covariance Test

Next, we trained an ensemble model to predict the outcome using only the covariates, without including the treatment variable (eligibility). This allowed us to estimate the part of the relationship between treatment and outcome that isn't explained by the covariates. Using this model along with our previously estimated propensity scores, we calculated the estimate $\hat{\theta} = 1,464$ (95% CI 1,061–1,867); see **Table 9**.

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n (A_i - \hat{p}_i)(Y_i - \hat{Q}_i)$$

$$IF_i = (A_i - \hat{p}_i)(Y_i - \hat{Q}_i) - \hat{\theta}$$

With a p value 1.07×10^{-12} , even under suboptimal bias correction or slight model misspecifications (residual confounding), we can cautiously reject the null hypothesis that treatment is not directly contributing to the outcome.

Conditional treatment effects using R-learner

An **R-learner** (without cross-fitting) produced conditional treatment-effect estimates. Because raw pseudo-outcomes were extreme, we bounded them at the 5th–95th percentiles before fitting; this trimming stabilized both the mean and the variance of the CATEs (Figure 4).

We observe a strong positive association between the estimated treatment effect with both age and income and a moderate effect on years of education. Since we did not include all covariates of the pension data set like prior financial assets, this pattern may reflect that individuals with higher incomes are likely to benefit more from 401(k) eligibility—potentially because higher income allows for greater contributions and compounding gains. Similarly, older individuals may exhibit larger estimated effects because they have had more time to accumulate assets within their retirement account. However, since we did not adjust for all possible confounders (like for example financial literacy), these associations should be interpreted with caution.

Conditional counterfactual prediction using DR-learner

We formed the pseudo-outcome

$$C_i = \hat{Y}_i^1 + \frac{A_i}{\hat{p}_i}(Y_i - \hat{Y}_i^1)$$

and regressed it **only** on age, income, and years of education; all other covariates are controlled through C_i . The resulting predictions $\hat{Y}^1$ represent the net financial assets if eligible for a 401k. Diagnostic plots are shown in **Figure 5**, and a summary performance metrics in **Table 10**.

We were required to bound the counterfactual outcome predictions from the DR learner as can be understood from the first histogram. This resulted in unusable predictions due to extreme outliers. Even with the bounded outcomes the Standard Error is unacceptably high. This may be attributed to the high variance and heavy tail of the measured outcomes as well as due to a lack of cross-fitting which leads to overfitting of our ensemble models.

Nevertheless we still see some defensible patterns in the predictions. Net financial assets for those who would have been eligible for a 401k are expected to rise with age and income. The main reason our smoothed spline for the income tapers down at the end is due to the applied bounding.

For education, we see an initial drop in predicted net financial assets at the lowest levels of schooling, likely reflecting individuals with very limited formal education and lower lifetime earnings potential. Beyond this early dip, assets tend to increase with more years of education, which aligns with expectations — higher education often leads to higher-paying jobs and better access to retirement plans like 401(k)s.

Conclusions

Our look at the 1991 SIPP data shows that just **being eligible for a 401(k)** is linked to **about \$8½ k more** in net financial assets on average (ATE), and **about \$10½ k more** for the people who were actually offered a plan (ATT). To be sure the manual calculations of the AIPW and TMLE were correct the results were compared and found to be in line with those obtained by performing a 1 fold ATE and ATT calculation using `npcausal`.

Notable issues that can negatively affect the quality of the analysis were:

- **Weak positivity.** Treated and untreated don't overlap much
- **We must adjust for income.** Income blocks the path *Eligibility - Income - Saving preference - Assets*. It is difficult to verify whether income sufficiently captures the bias introduced due to unmeasured saving preferences.
- **No cross-fitting.** We trained the SuperLearner models on the same data we used for the effect, which can overfit and push standard errors up. A simple K-fold split would fix this.
- **Very spread-out outcome.** Net assets run from −\$500 k to +\$1.5 M. Trimming helps, but the big (mostly one sided) range still inflates the standard error and the trimming has the potential to obfuscate important information.

Even with these limits, every method points the same way: workers who can join a 401(k) accumulate more financial wealth than similar workers who can't.

Appendix

Term	Estimate	95% CI Lower	95% CI Upper
Treatment A	19559.3	16999.9	22118.8

Table 1. Unadjusted OLS regression of net_tfa on e401 .

Outcome learner	Value
SL.glm.interaction_All	0.2975992
SL.ranger_All	0.0990618
SL.earth_All	0.6033390

Propensity score learner	Weight
SL.glmnet_All	0.0464
SL.glm.interaction_All	0.1741
SL.mean_All	0.0001
SL.ranger_All	0.1928
SL.earth_All	0.5866

Table 2 and 3. model weights obtained for outcome and propensity score prediction.

Metric	Value
ATE (AIPW estimate)	8,761.44
Standard Error	973.80
Mean Influence Function	≈ 0
95% CI Lower Bound	6,852.79
95% CI Upper Bound	10,670.10

Table 4. AIPW estimate of the average treatment effect (ATE).

Metric	Value
ATE (TMLE estimate)	8,481.22
Standard Error	973.79
Mean Influence Function	≈ 0
95% CI Lower Bound	6,572.59
95% CI Upper Bound	10,389.85

Table 5. TMLE estimate of the average treatment effect (ATE).



Statistic	Value
Min	-502,302
1st Quartile (Q1)	-500
Median	1,499
Mean	18,052
3rd Quartile (Q3)	16,524
Max	1,536,798

Figure 2 and Table 6. Extreme variability and fat-tail in the outcome distribution.

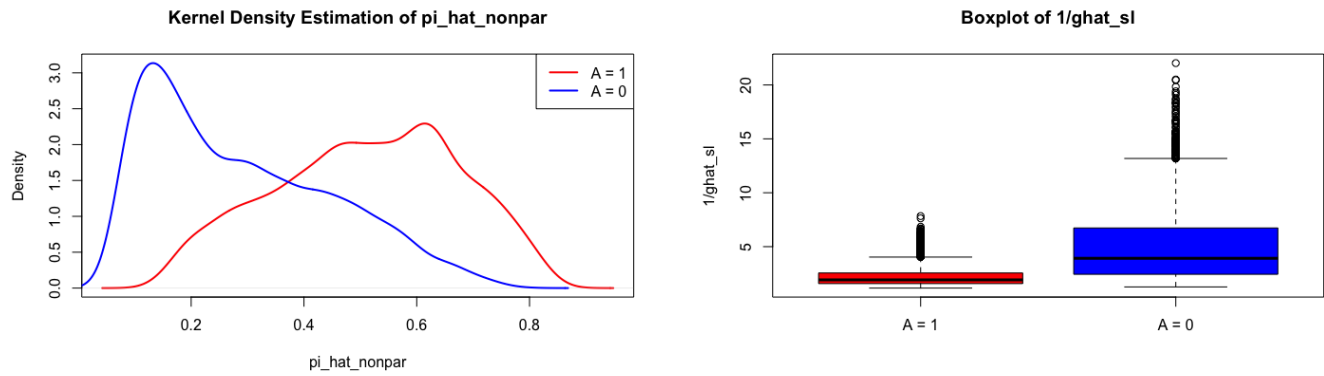


Figure 3. Violation of positivity assumption and large inverse probability weights

Metric	Value
ATT (AIPW estimate)	10,548.08
Standard Error	1,395.24
Mean Influence Function	≈ 0
95% CI Lower Bound	7,813.42
95% CI Upper Bound	13,282.74

Table 7. AIPW estimate of the average treatment effect on the treated (ATT).

Metric	Value
ATT (TMLE estimate)	10,467.70
Standard Error	1,401.95
Mean Influence Function	≈ 0
95% CI Lower Bound	7,719.87
95% CI Upper Bound	13,215.52

Table 8. TMLE estimate of the average treatment effect on the treated (ATT).

Metric	Value
Estimate (θ)	1,464.24
Standard Error	205.61
95% CI Lower Bound	1,061.23
95% CI Upper Bound	1,867.24
p-value	1.07×10^{-12}

Table 9. Residual covariance test results.

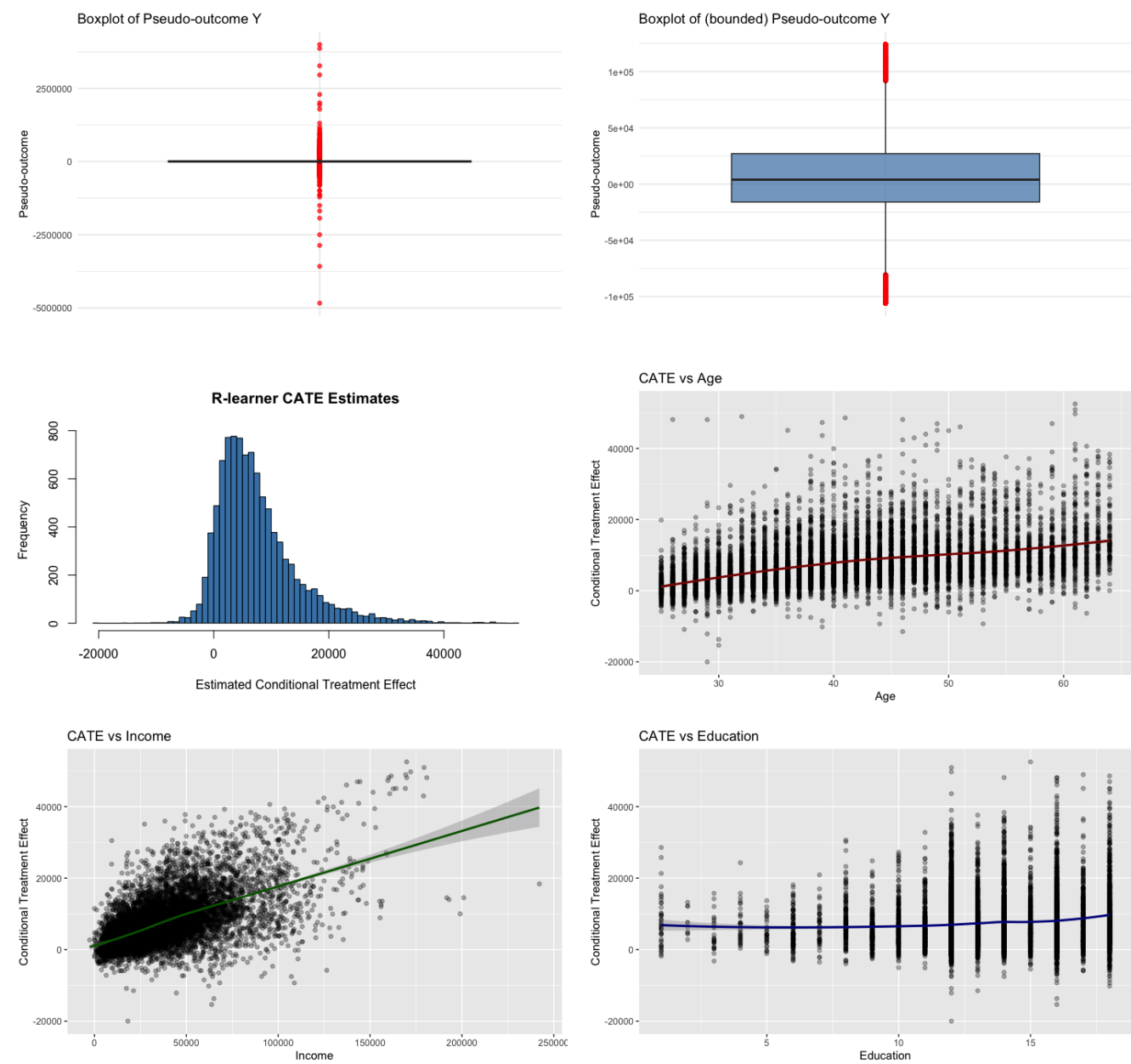


Figure 4. Conditional treatment effects obtained by an R-learner with respect to age, income and education.

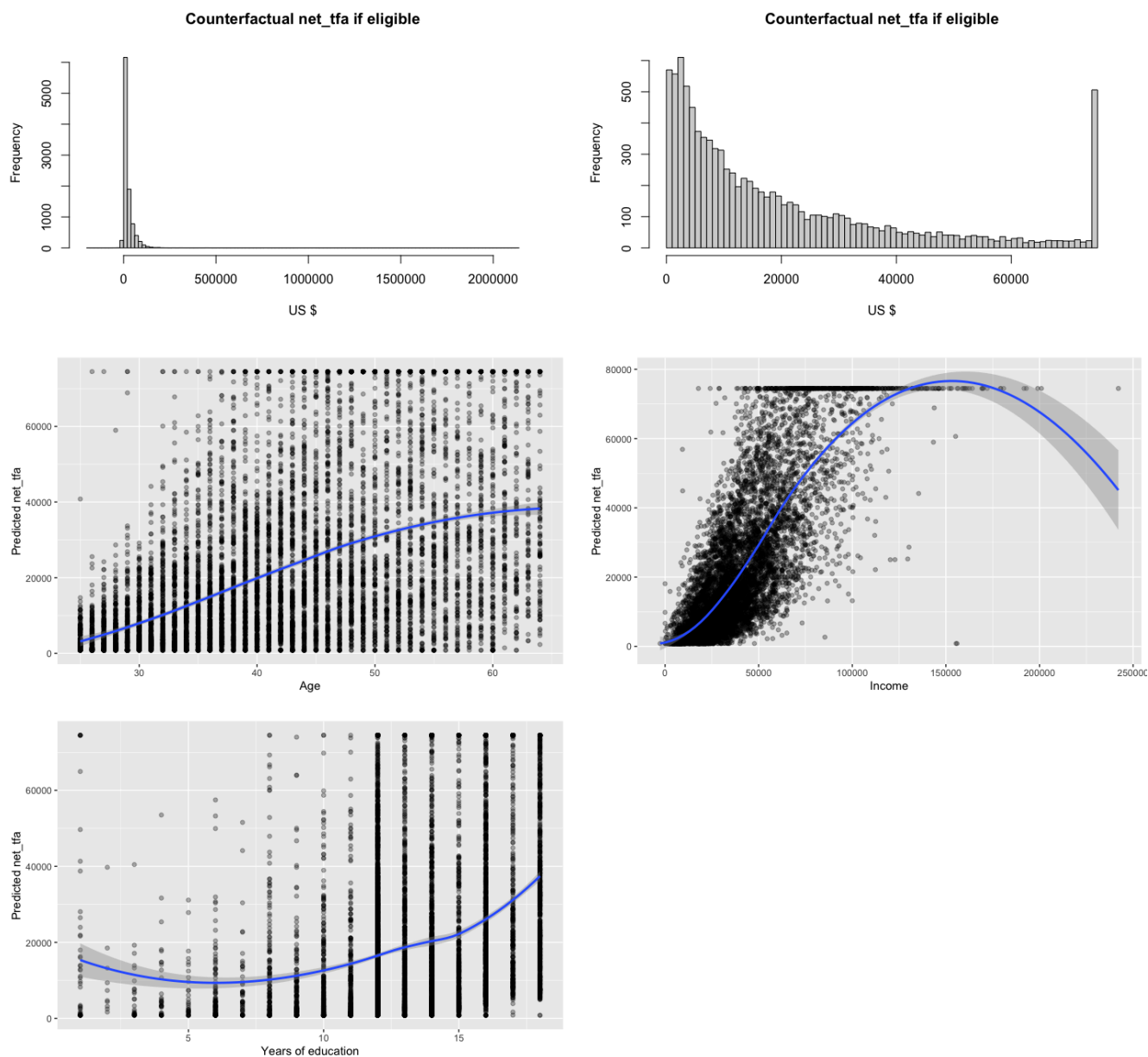


Figure 5. Histograms for unbounded and bounded counterfactual outcome predictions, as well as plots of counterfactual outcome prediction variation by age, income and years of education.

Metric	Value
MSE	4,371,106,405
SE	1,273,188,010
CI Lower	1,875,657,906
CI Upper	6,866,554,903

Table 10. Performance metrics for the DR-learner counterfactual predictions.